Southern New Hampshire University

Storm and Climate Data Record (SCDR)

Implementation Plan

Date of record: 12/14/2025

Calendar Year 2024

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I. Introduction

1.1 Overview and purpose

The purpose of this project is to help the police department examine whether there is a higher percentage of property crime in relation to storm occurences. This report will help the police predict possible future crimes. The data being analyzed will include crimes that occurred in the city of Miami during storm events in the year 2024.

1.2 Define why you need data analysis

This data analysis is needed to determine if there is an increased property crime count during storm events. This will help the police department predict crimes before they occur. These predictions can help to adjust the police department's resource allocation, strategy, and response times during weather events. These strategies can help discourage future crime

II. Data Preparations

2.1 Name data sources

The crime data is stored as a table in "mpdCrimeData2024". The public crime data was provided by the City of Miami Police Department. The storm data is stored as a table in "stormEvents2024". The data was collected from historical weather and storm event data from the National Centers for Environmental Information, a division of NOAA.

2.2 Filter through unnecessary data

There are several variables available in both data sets but only a few variables are necessary to filter through unnecessary data. In the mpdCrimeData2024 table I used the CFSDate, IncidentType, and ZIP Code columns.This allowed me to define the date, set for property type crimes only, and search for these crimes in Miami-Dade. In the stormevents2024 table I used the BEGIN_YEARMONTH, BEGIN_DAY, CZ_FIPS, and EVENT_Type columns.

These columns allowed me to reformat the date column so it would be the same as the crime table, set the weather events to occurances in Miaimi_Dade, and check to see if storms occurred on which days in 2024. These columns allowed to create the final merged table that included the same formatted date from 2024-01-01 to 2024-12-31, number of property crime incidents in Miami-Dade organized by day of occurence, and number of storm incidents in Miami_Dade by day of occurence.

2.3 Define your parameters

The parameters used in this data analysis include the dates of which property crimes occurred during stormy and nonstormy days throughout 2024 in Miami-Dade Florida. This data was used to create a two line graph. The x axis lists the months in 2024 to compare the crime data over time. The y axis represents the total number of property crime incidents. The lines will be representing the stormy and nonstormy datatables to compare property crime incidents during these variables.

2.4 Identify measurement priorities

The measurement priorities of this data analysis is to measure the property crime incidents during stormy and nonstormy days. This way the data can produce a two-sample test to see if the difference in weather is statistically significant in comparison to property crime incidents.

2.5 Ensure collected data fits the need

The need for this data analysis is to be able to determine whether there is a statistical significance between property crime incident counts increasing during stormy weather in 2024 in Miami-Dade Florida. In order to ensure the collected data fits this need each of the datasets tracks each day throughout 2024 counting the total number of crime incidents and storm

incidents. We can then separate this data by the the number of storm incidents. If the count is zero we know those crimes on nonstormy and if the count is greater than zero we know the crimes occurred on stormy days.

III. Data Analysis

3.1 Identify scripts used

The script used to create the final table of property crime incidents on non stormy days for everyday in 2024 in Miami-Dade:

```
#table that contains property crime counts on non stormy days
no_storms=finalmerge_df[finalmerge_df['storm_counts']==0]
print(no_storms)
```

The script used to perform preliminary data analysis on property crime incidents on nonstormy days:

```
#statistics for property crime counts on non storm days
stats_no_storm = no_storms['property_counts'].describe()
print(f'STATS for property crimes during NO storms: \n{stats_no_storm}')
print()
```

The script used to create the final table of property crime incidents on stormy days for everyday in 2024 in Miami-Dade:

```
#table that contains property crime counts on stormy days
yes_storms=finalmerge_df[finalmerge_df['storm_counts'] > 0]
print(yes_storms)
```

The script used to perform preliminary data analysis on property crime incidents on stormy days:

```
#statistics for property crime counts on stormy days
stats_yes_storm=yes_storms['property_counts'].describe()
print(f'STATS for property crimes during YES storms: \n{stats_yes_storm}')
print()
```

The script used to identify in a two-sample test whether there is statistical signifincance on storms affecting property crime incidents:

```
#finding statistical significance
t_statistic, p_value = stats.ttest_ind(no_storms['property_counts'], yes_storms['property_counts'], nan_policy='omit')
print(f"T-statistic: {t_statistic}")
print(f"P-value: {p_value}")
```

The script used to create the two line time graph depicting property crime incidents on stormy versus nonstormy day in 2024 in Miami-Dade:

```
#creating line graph to compare property crime counts during stormy and nonstormy weather
ax = no_storms.plot(x='stormDay', y='property_counts', label ='no storms', kind='line', figsize=(12,6))
yes_storms.plot(ax=ax, x='stormDay', y='property_counts', label = 'yes storms', kind='line')

ax.set_title('Property Crimes Throughout 2024 to evaluate change due to Weather')
ax.set_xlabel('Dates throughout 2024')
ax.set_ylabel('Property Crime Counts')
ax.legend()
plt.grid(True)
plt.show()
```

3.2 Run the scripts to analyze the data and validate the output

The resulting script from table creation to track property crime incidents on nonstormy days is below.

```
      stormDay  storm_counts  property_counts
1   2024-01-02           0.0              901
2   2024-01-03           0.0              970
5   2024-01-06           0.0              765
6   2024-01-07           0.0              660
9   2024-01-10           0.0              986
..         ...           ...              ...
359 2024-12-25           0.0              501
360 2024-12-26           0.0              723
361 2024-12-27           0.0              763
364 2024-12-30           0.0              731
365 2024-12-31           0.0              687

[245 rows x 3 columns]
```

The resulting script from prelimary analysis of property counts on nonstormy days is below.

```
STATS for property crimes during NO storms:
count     245.000000
mean      853.342857
std       101.210323
min       501.000000
25%       784.000000
50%       854.000000
75%       928.000000
max      1134.000000
Name: property_counts, dtype: float64
```

The resulting script from table creation to track property crime incidents on stormy days is below.

```
      stormDay  storm_counts  property_counts
0   2024-01-01           3.0              895
3   2024-01-04           1.0              932
4   2024-01-05           1.0              794
7   2024-01-08           8.0              853
8   2024-01-09           3.0              894
..         ...           ...              ...
345 2024-12-11           1.0              898
348 2024-12-14           2.0              768
353 2024-12-19           1.0              873
362 2024-12-28           1.0              673
363 2024-12-29           2.0              614

[121 rows x 3 columns]
```

The resulting script from preliminary analysis of property counts on stormy days is below.

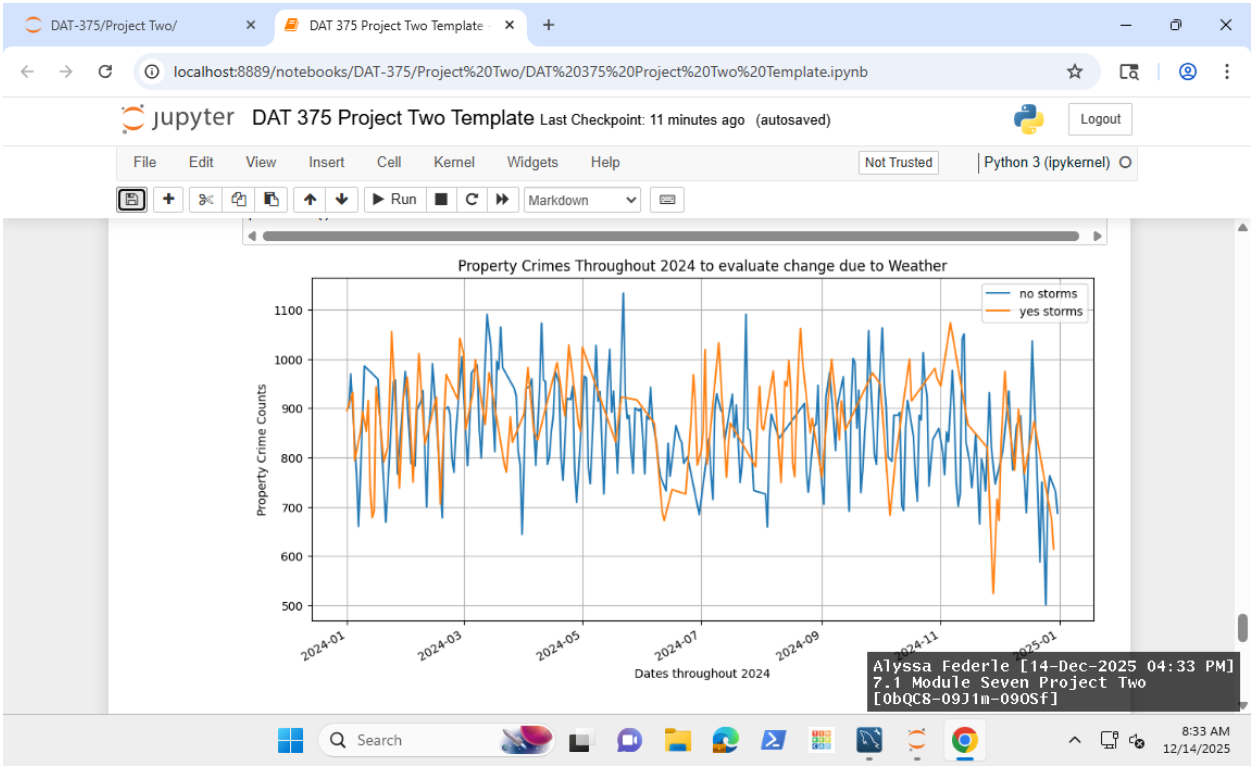
```
STATS for property crimes during YES storms:
count     121.000000
mean      870.206612
std       104.896848
min       524.000000
25%       805.000000
50%       873.000000
75%       945.000000
max      1074.000000
Name: property_counts, dtype: float64
```

The resulting script from the two-sample statististical significance test is below.

```
T-statistic: -1.4815575436261954
P-value: 0.1393231895959585
```

IV. Drawing Conclusions

4.1 Present the results of the analysis to stakeholders



The visual graph is shown above. The x-axis shows the dates throughout 2024 and is separated on a monthly basis. The yaxis tracks the total number of property crime incidents. The legend in the corner explains that the blue line is tracking the nonstormy days while the orange line is tracking stormy days. There also grid lines on the graph to make the graph more easily read. As you can see from the graph the lines are majorily overalapping.

4.2 Determine whether the problem was addressed, including any challenges and limitations

This problem was addressed indicating that there is not a strong statistical significance of an increase of property crime incidents on stormy days. There could be challenges in the type of stormy weather crimes as my analysis included all storm types. Further analysis could be done to see if certain weather incidents increase crime. Some zip codes may also be affecting the data as they may be more crime ridden than others.

4.3 Report potential new findings

Based on these finding I would have to conclude that there is not a statistical relationship between increased property crime incidents and stormy weather. The lines in the graph are more less overlapping and a lack a significant visual difference. The p-value from the two-sample data test is also over 5% which indicates no significant statistical relationship.

References

Crime mapping - miami police department. (n.d.). https://www.miami-police.org/CrimeMapping.html

FBI — crime in the U.S. 2019. (n.d.-a). https://ucr.fbi.gov/crime-in-the-u.s/2019/crime-in-the-u.s.-2019

Ncei. (n.d.). *Storm events database*. National Centers for Environmental Information. https://www.ncdc.noaa.gov/stormevents/

Understanding T-statistic, F-statistic, and P-value in statistical analysis | by Kishan A | medium. (n.d.-c). https://kishanakbari.medium.com/understanding-t-statistic-f-statistic-and-p-value-in-statistical-analysis-562463988bdc